Linear Algebra

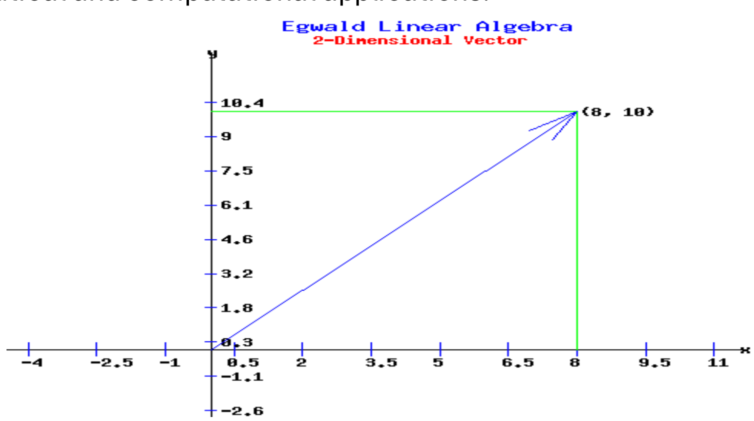
Linear algebra is a branch of mathematics that deals with vectors, matrices, and systems of linear equations. It focuses on understanding how these mathematical structures interact and transform through operations like addition, multiplication, and inversion.

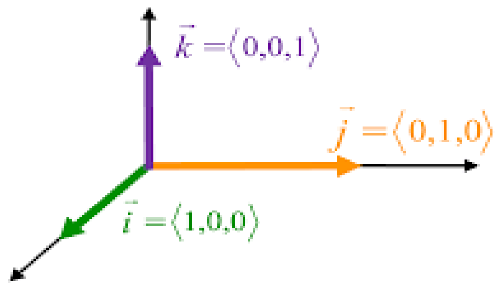
Linear algebra is essential in data science as it helps in organizing, processing, and analyzing large datasets efficiently. Many datasets are represented as matrices, where each row corresponds to a data sample and each column represents a feature. Techniques like Principal Component Analysis (PCA) use linear algebra to reduce the number of features while preserving important information, making computations faster and improving model performance. Machine learning algorithms, including linear regression, support vector machines (SVM), and neural networks, rely on matrix operations for training and optimization. In image processing, images are stored as matrices of pixel values, and linear algebra is used for filtering, transformations, and compression through methods like Singular Value Decomposition (SVD). Recommendation systems, such as those used by Netflix and Spotify, apply matrix factorization to analyze user preferences and provide personalized content. Overall, linear algebra is a foundational tool in data science, enabling efficient data manipulation, pattern recognition, and predictive modeling.

Vectors in Linear Algebra

In linear algebra, **vectors** are mathematical objects that have both **magnitude** (size) and **direction**. They are often represented as ordered lists of numbers, called components or coordinates, which define their position in space. Vectors can exist in two-dimensional (2D), three-dimensional (3D), or even higher-dimensional spaces.

Vectors are used in many applications, such as physics (to represent forces and velocities), computer graphics (to define positions and movements), and machine learning (to represent data points). A simple example of a vector in **2D space** is **v = (3, 4),** where 3 represents the movement in the x-direction, and 4 represents the movement in the y-direction. Similarly, in **3D space**, a vector could be **v = (2, -1, 5),** where the three numbers represent movements along the x, y, and z axes, respectively. Vectors can be added, subtracted, and multiplied by a scalar, making them useful for many mathematical and computational applications.





Vector space and vector space in a data set

A **vector space** in linear algebra is a collection of vectors that can be added together and multiplied by scalars (numbers) while remaining in the same space. It follows certain mathematical rules, such as associativity, commutativity, and distributivity. Vector spaces provide the foundation for solving linear equations, transformations, and many real-world applications.

Example of a Vector Space:
Consider a 2D vector space, where vectors like **(2,3)** and **(1, -1)** exist. If we add them, we get **(3,2)**, which is still within the same space. Similarly, multiplying **(2,3)** by 2 gives **(4,6)**, which is also in the vector space. This property holds for higher dimensions as well.

Vector Space in a Dataset:
In data science, a **vector space** represents data points as vectors in a multi-dimensional space, where each feature of the dataset corresponds to a dimension. For example, if a data set has three features (e.g., height, weight, and age), each data point can be represented as a 3D vector **(h, w, a)**. Machine learning algorithms, such as clustering and classification, operate within this vector space by finding relationships, distances, and patterns among data points.

Example of a Vector Space in a Dataset
Consider a dataset of students with three features: **age, height, and weight**. Each student can be represented as a vector in a 3D vector space, where:

- **Age (years)** → First dimension (x-axis)
- **Height (cm)** → Second dimension (y-axis)
- **Weight (kg)** → Third dimension (z-axis)

Example Data Points (Vectors):

1. Student A → **(18, 170, 65)**
2. Student B → **(20, 175, 70)**
3. Student C → **(22, 160, 55)**

Each student's data point is a vector in a 3D space. In machine learning, these vectors can be used for clustering (e.g., grouping students with similar physical traits), classification (e.g., predicting health conditions), or regression (e.g., estimating weight based on age and height). This concept is fundamental in **feature space representations**, where each feature defines a new axis in a higher-dimensional vector space.

Matrices

Definition of Matrices

A **matrix** is a rectangular array of numbers, symbols, or expressions arranged in rows and columns. It is used to represent and manipulate data in various fields like computer science, physics, engineering, and economics. A matrix is usually denoted by a capital letter (e.g., **A, B, C**) and enclosed in brackets:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Here, matrix **A** has **3 rows** and **3 columns**, making it a **3 × 3 matrix**.

Entries in a Matrix

Each element inside a matrix is called an **entry** and is usually represented as aij, where:

- i is the **row number**
- j is the **column number**

For example, in the matrix:

$$B = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

- $b_{11} = 2$ (1st row, 1st column)
- $b_{12} = 4$ (1st row, 2nd column)
- $b_{21} = 6$ (2nd row, 1st column)
- $b_{22} = 8$ (2nd row, 2nd column)

Arithmetic Operations on Matrices

1. Matrix Addition

Two matrices of the same size can be added by adding their corresponding elements.

Example:
If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Then,

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

2. Matrix Subtraction

Two matrices of the same size can be subtracted by subtracting their corresponding elements.

Example:

$$A - B = \begin{bmatrix} 1-5 & 2-6 \\ 3-7 & 4-8 \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

3. Scalar Multiplication

A matrix can be multiplied by a scalar (a single number) by multiplying each element by the scalar.

Example:
If

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

and scalar $k = 3$, then

$$3A = \begin{bmatrix} 3 \times 2 & 3 \times 4 \\ 3 \times 6 & 3 \times 8 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 18 & 24 \end{bmatrix}$$

4. Matrix Multiplication

Matrix multiplication is **not element-wise**; it follows row-column multiplication. If **A** is an $m \times n$ matrix and **B** is an $n \times p$ matrix, their product **AB** is an $m \times p$ matrix.

Example:

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Then,

$$AB = \begin{bmatrix} (1 \times 5 + 2 \times 7) & (1 \times 6 + 2 \times 8) \\ (3 \times 5 + 4 \times 7) & (3 \times 6 + 4 \times 8) \end{bmatrix}$$

$$= \begin{bmatrix} (5 + 14) & (6 + 16) \\ (15 + 28) & (18 + 32) \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Types of Matrices with Examples

Matrices can be classified based on their structure, size, and properties. Below are the main types of matrices along with examples:

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1. Row Matrix

A matrix that has only **one row** is called a **row matrix**.

✅ **Example:**

$$R = \begin{bmatrix} 3 & 5 & 7 \end{bmatrix}$$

(1×3 matrix)

2. Column Matrix

A matrix that has only **one column** is called a **column matrix**.

✅ **Example:**

$$C = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}$$

(3×1 matrix)

3. Square Matrix

A matrix where the **number of rows is equal to the number of columns**.

✅ **Example:**

$$S = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(3 × 3 matrix)

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4. Rectangular Matrix

A matrix where the **number of rows is not equal to the number of columns**.

✅ **Example:**

$$R = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

(2 × 3 matrix)

5. Diagonal Matrix

A square matrix where **all non-diagonal elements are zero**.

✅ **Example:**

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

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6. Scalar Matrix

A diagonal matrix where **all diagonal elements are the same**.

✅ **Example:**

$$S = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

7. Identity Matrix (Unit Matrix)

A square matrix where **all diagonal elements are 1, and the rest are 0.**

✅ **Example:**

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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8. Zero Matrix (Null Matrix)

A matrix where **all elements are zero.**

✅ **Example:**

$$Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

9. Upper Triangular Matrix

A square matrix where **all elements below the main diagonal are zero.**

✅ **Example:**

$$U = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix}$$

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10. Lower Triangular Matrix

A square matrix where **all elements above the main diagonal are zero.**

✅ **Example:**

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 6 & 7 \end{bmatrix}$$

11. Symmetric Matrix

A square matrix where $A^T = A$ **(the matrix is equal to its transpose)**.

✅ **Example:**

$$S = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

12. Skew-Symmetric Matrix

A square matrix where $A^T = -A$ **(the transpose of the matrix is its negative)**.

✅ **Example:**

$$S = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 0 & -5 \\ 3 & 5 & 0 \end{bmatrix}$$

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13. Orthogonal Matrix

A square matrix where $A^T \cdot A = I$ **(the product of the matrix and its transpose is the identity matrix)**.

✅ **Example:**

$$O = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

14. Singular Matrix

A square matrix whose **determinant is zero**.

✅ **Example:**

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\text{Determinant} = (2 \times 2) - (4 \times 1) = 4 - 4 = 0$$

Since the determinant is **zero**, it is a **singular matrix**.

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15. Non-Singular Matrix

A square matrix whose **determinant is non-zero**.

✅ **Example:**

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$$

$$\text{Determinant} = (3 \times 2) - (1 \times 4) = 6 - 4 = 2$$

Since the determinant is **non-zero**, it is a **non-singular matrix**.

16. Idempotent Matrix

A matrix that satisfies $A^2 = A$ (multiplying it by itself gives the same matrix).

✅ **Example:**

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $A^2 = A$, it is an **idempotent matrix**.

17. Involutory Matrix

A matrix that satisfies $A^2 = I$ (multiplying it by itself gives the identity matrix).

✅ **Example:**

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Since $A^2 = I$, it is an **involutory matrix**.

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18. Nilpotent Matrix

A matrix where $A^n = 0$ **for some integer** n.

✅ **Example:**

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Since $A^2 = 0$, it is a **nilpotent matrix**.

Cartesian coordinates

Definition

Cartesian coordinates are a system used to locate points in a space using numerical values. Each point is defined by its distance from perpendicular reference lines called **axes**. This system is widely used in mathematics, physics, engineering, and computer graphics.

X and Y Coordinates

In a **2D Cartesian coordinate system**, a point is represented as **(x, y)** where:

- **x-coordinate** (abscissa) represents the horizontal position.
- **y-coordinate** (ordinate) represents the vertical position.

Example:

- The point **(3, 4)** means:
  - Move **3 units right** on the **x-axis**.

- Move **4 units up** on the **y-axis**.

2D Cartesian Coordinate System

In **two-dimensional (2D) space**, the Cartesian plane consists of:

- **X-axis (horizontal)**
- **Y-axis (vertical)**
- **Origin (0,0)** – The intersection of both axes.

Example:

- (5,2) → 5 steps right, 2 steps up.
- (-3, -4) → 3 steps left, 4 steps down.

Uses:

- Graphing equations
- Mapping locations
- Designing computer graphics

3D Cartesian Coordinate System

In **three-dimensional (3D) space**, a point is represented as **(x, y, z)** where:

- **x-coordinate** – Left or right position.
- **y-coordinate** – Up or down position.
- **z-coordinate** – Forward or backward position.

Example:

- Points **(3, 4, 5)** means:
  - Move **3 units along x-axis**.
  - Move **4 units along y-axis**.
  - Move **5 units along z-axis** (depth).

Uses:

- 3D modeling in games
- Physics simulations
- Robotics and engineering

Unit Vectors and Scalers of Vectors

1. Unit Vector

A **unit vector** is a vector with a **magnitude (length) of 1**. It is used to indicate direction without changing the magnitude.

✅ **Formula for a Unit Vector**:

$$\hat{v} = \frac{v}{|v|}$$

where:

- $\hat{v}$ = unit vector
- v = given vector
- $|v|$ = magnitude of the vector

✅ **Example:**
Given vector $v = (3, 4)$, its magnitude is:

$$|v| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

So, the unit vector is:

$$\hat{v} = \left(\frac{3}{5}, \frac{4}{5}\right)$$

Uses of Unit Vectors:

- Representing direction in physics and engineering
- Normalizing vectors in machine learning
- Defining coordinate axes (e.g., $\hat{i}$, $\hat{j}$, $\hat{k}$ in 3D)

2. Scalars of Vectors

A **scalar** is a quantity that has only **magnitude** but no direction. Scalars are used to **scale a vector** by multiplying it.

✅ **Examples of Scalars:**

- **Speed** (not velocity)
- **Mass**
- **Temperature**
- **Time**

✅ **Example of Scalar Multiplication:**
Let vector $v = (2, 3)$ and scalar $k = 4$:

$$4 \cdot (2, 3) = (8, 12)$$

This means the vector's direction remains the same, but its magnitude is increased by a factor of 4.

Uses of Scalars in Vectors:

- Changing the size of a force or velocity in physics

- Rescaling vectors in 3D modeling
- Adjusting brightness in image processing

Vector Addition

Definition:

Vector addition is the process of combining two or more vectors to form a resultant vector. This operation follows the **parallelogram law** or the **triangle law** of vector addition.

If **A** and **B** are two vectors, their sum is:

R=A+B

where **R** is the resultant vector.

Methods of Vector Addition

1. Triangle Law of Vector Addition

- Place the tail of **B** at the head of **A**.
- The resultant **R** is the vector from the tail of **A** to the head of **B**.

Example:

If **A= (3,2)** and **B= (1,4)**, then:

R=A+B

= (3+1,2+4)

= (4,6).

2. Parallelogram Law of Vector Addition

- Draw both vectors from a common point.
- From a parallelogram with these vectors as adjacent sides.
- The diagonal represents the resultant vector.

Example:

Given **A= (2,3)** and **B= (4,1)** their result is:

R= (2+4,3+1)

= (6,4)

Properties of Vector Addition:

1. **Commutative Law:** A+B=B+A
2. **Associative Law:** (A+B) +C=A+(B+C)

3. **Zero Vector:** A+0=A

X and Y intercept in Liner Algebra

In linear algebra, the **x-intercept** and **y-intercept** represent the points where a line crosses the **x-axis** and **y-axis**, respectively.

1. X-Intercept

The **x-intercept** is the point where the line crosses the **x-axis**. At this point, the **y-coordinate is zero** (y=0y = 0).

To find the x-intercept:

- Set y=0 in the equation of the line and solve for xx.

Example:

Given the line equation:

2x+3y=6

Set y=0:

2x+3(0) =6

So, the **x-intercept is (3,0)**.

2. Y-Intercept

The **y-intercept** is the point where the line crosses the **y-axis**. At this point, the **x-coordinate is zero** (x=0).

To find the y-intercept:

- Set x=0 in the equation of the line and solve for yy.

Example:

Given the same equation:

2x+3y= 6

Set x=0:

2(0) +3y= 6

So, the **y-intercept is (0,2)**.

Graphical Interpretation

- The **x-intercept** is where the line meets the x-axis.
- The **y-intercept** is where the line meets the y-axis.
- The general equation of a line is **y=mx+c** where:
    - m is the slope

- c is the y-intercept

Dot and Cross product of Vectors

Vectors are mathematical entities that have both **magnitude** and **direction**. Two important operations involving vectors are the **dot product** (also called the scalar product) and the **cross product** (also called the vector product). Let's explore both in detail.

Dot Product of Two Vectors

The **dot product** of two vectors results in a **scalar** quantity (a single number).
Formula:
For two vectors, **A** and **B**:
$\mathbf{A \cdot B = |A||B| \cos\theta}$
where:

- |A| and |B| are the magnitudes of vectors **A** and **B**
- θ is the angle between the two vectors

If the vectors are given in component form:
$A = (A_x, A_y, A_z)$,
$B = (B_x, B_y, B_z)$
Then, the dot product is:
$A \cdot B = A_x * B_x + A_y * B_y + A_z * B_z$.

Example:
A= (3,4,5),
B= (2, −1,3)
A·B= (3)(2) +(4) (−1) +(5)(3)
=6−4+15
=17.

Cross Product of Two Vectors

The **cross product** of two vectors results in a **new vector** that is **perpendicular** to both.
Formula:
For two vectors, **A** and **B**:
$A \times B = |A||B| \sin\theta n$

where:

- |A| and |B| are the magnitudes of vectors **A** and **B**.
- θ is the angle between the vectors.
- n is a unit vector **perpendicular** to both **A** and **B** (following the right-hand rule).

In component form, if:

$$\mathbf{A} = (A_x, A_y, A_z), \quad \mathbf{B} = (B_x, B_y, B_z)$$

Then the cross product is given by the determinant:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Expanding the determinant:

$$\mathbf{A} \times \mathbf{B} = \mathbf{i}(A_y B_z - A_z B_y) - \mathbf{j}(A_x B_z - A_z B_x) + \mathbf{k}(A_x B_y - A_y B_x)$$

Properties of Cross Product:

1. **Not Commutative:** A×B=−(B×A).
2. **Distributive:** A×(B+C) =A×B+A×C).
3. **Parallel Vectors:** If A×B=0, the vectors are **parallel** or **collinear**.

Example:

Let:

$$\mathbf{A} = (3, -3, 1), \quad \mathbf{B} = (4, 9, 2)$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -3 & 1 \\ 4 & 9 & 2 \end{vmatrix}$$

Expanding:

$$= \mathbf{i}((-3)(2) - (1)(9)) - \mathbf{j}((3)(2) - (1)(4)) + \mathbf{k}((3)(9) - (-3)(4))$$

$$= \mathbf{i}(-6 - 9) - \mathbf{j}(6 - 4) + \mathbf{k}(27 + 12)$$

$$= \mathbf{i}(-15) - \mathbf{j}(2) + \mathbf{k}(39)$$

$$= (-15, -2, 39)$$

So, the **cross product** $\mathbf{A} \times \mathbf{B} = (-15, -2, 39)$.

Basic Vectors and Transformations

1. Basics of Vectors

A **vector** is a mathematical quantity that has both **magnitude** (size) and **direction**. It is different from a scalar, which only has magnitude.

Types of Vectors:

1. **Zero Vector (Null Vector):** A vector with zero magnitude, denoted as **(0,0)** in 2D or **(0,0,0)** in 3D.
2. **Unit Vector:** A vector with a magnitude of **1**, used to indicate direction.
3. **Position Vector:** A vector that represents the position of a point with respect to an origin.
4. **Equal Vectors:** Two vectors are equal if they have the same magnitude and direction.
5. **Negative Vector:** A vector that has the same magnitude as another vector but in the opposite direction.
6. **Collinear Vectors:** Vectors that are parallel or lie along the same line.
7. **Co-planar Vectors:** Vectors that lie in the same plane.

Representation of a Vector

A vector **A** is typically represented as:

$$\mathbf{A} = (A_x, A_y) \quad \text{(in 2D)}$$

$$\mathbf{A} = (A_x, A_y, A_z) \quad \text{(in 3D)}$$

where:

- A_x, A_y, and A_z are the components along the **x**, **y**, and **z** axes, respectively.

Example:

If a vector moves 3 units in the **x-direction** and 4 units in the **y-direction**, it is represented as:

$$\mathbf{A} = (3, 4)$$

The **magnitude** of this vector is:

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

2. <u>Vector Transformations</u>

Vector transformations involve changing the vector's position, orientation, or size while preserving its properties.

Types of Vector Transformations:

A. Translation (Shifting)

A vector is moved **without** changing its direction or length.

Example: If vector **A = (3,4)** is translated by vector **B = (1,2)**, the new vector **A'** will be:

A'= (Ax+Bx, Ay+By)
= (3+1,4+2)
= (4,6)

B. Scaling (Change in Magnitude)

A vector is **stretched** or **shrunk** by a scalar factor **k**.

Example: If k=2k = 2k=2 and **A = (3,4)**, then the scaled vector is:

A'= (2×3,2×4)
= (6,8)
Scaling up (k > 1): Vector increases in magnitude.
Scaling down (0 < k < 1): Vector shrinks.
Negative scaling (k < 0): Vector flips direction.

C. Rotation

A vector is rotated by an angle **θ** around the origin.

If **A = (x, y)** is rotated by an angle **θ**, the new vector A' is:

$A' = (x', y') = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$
Example:
Rotating vector **(3,4)** by **90°** counterclockwise:
$A' = (3\cos 90^\circ - 4\sin 90^\circ, 3\sin 90^\circ + 4\cos 90^\circ)$

A'=(3cos90∘−4sin90∘,3sin90∘+4cos90∘)
= (3(0) −4(1),3(1) +4(0))
= (−4,3)

D. Reflection

A vector flipped over an axis.

Reflection over the x-axis:

A'= (x, −y)

A' = (x, -y)

A'= (x, −y)

Reflection over the y-axis:

A'= (−x, y)

A' = (-x, y)

A'= (−x, y)

Example: Reflecting **(3,4)** over the x-axis:

A'= (3, −4)

A' = (3, -4)

A'= (3, −4)

Liner Transformation and Matrices

1. Linear Transformation

A **linear transformation** is a function that maps one vector space to another while preserving vector addition and scalar multiplication.

Definition:

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if, for all vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ and any scalar c:

1. **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. **Homogeneity:** $T(c\mathbf{u}) = cT(\mathbf{u})$

Examples of Linear Transformations:

1. **Scaling:** $T(x, y) = (2x, 3y)$
2. **Reflection over x-axis:** $T(x, y) = (x, -y)$
3. **Rotation by** θ:

$$T(x, y) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$$

4. **Shear Transformation:**
   - Horizontal Shear: $T(x, y) = (x + ky, y)$
   - Vertical Shear: $T(x, y) = (x, y + kx)$

Properties of Linear Transformations

- The origin **(0,0)** always maps to itself.
- Straight lines remain straight.
- Parallel lines remain parallel.

2. Matrices and Linear Transformations

Every linear transformation can be represented using a **matrix**.

Matrix Representation of a Linear Transformation

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be written as:

$$T(\mathbf{x}) = A\mathbf{x}$$

where:

- A is an $m \times n$ transformation matrix.
- $\mathbf{x}$ is a column vector.

Examples:

1. Scaling Transformation

Scaling by factors a and b:

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

For $a = 2, b = 3$:

For $a = 2, b = 3$:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

2. Rotation Matrix

Rotating a vector by θ:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

For $\theta = 90°$:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

3. Reflection Across the X-axis

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

4. Shear Transformation (Horizontal)

$$A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

where k controls the shear factor.

3. Matrix Transformations in 3D

For 3D transformations, the matrix size is 3×3.

Example: Rotation about the Z-axis

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Composite Transformations

Multiple transformations can be **combined** using matrix multiplication:

$$T = A_2 A_1$$

where:

- A_1 applies the first transformation.
- A_2 applies the second transformation.

Example: **Rotation followed by Scaling**

$$T = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Shear Transformation and Why do we need Transformation

1. What is Shear Transformation?

A **shear transformation** is a type of linear transformation that **distorts the shape** of an object **without changing its area**. It shifts points in a specific direction **parallel to an axis** but **does not change their perpendicular distance** from that axis.

Types of Shear Transformations:

1. **Horizontal Shear (X-Shear)** – Moves points **horizontally**, keeping their **y-coordinates unchanged**.
2. **Vertical Shear (Y-Shear)** – Moves points **vertically**, keeping their **x-coordinates unchanged**.

Mathematical Representation:

The shear transformation is represented using a **matrix.**

A. Horizontal Shear:

$$A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

- k is the shear factor.
- A point (x, y) transforms as:

$$x' = x + ky, \quad y' = y$$

B. Vertical Shear:

$$A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

- A point (x, y) transforms as:

$$x' = x, \quad y' = y + kx$$

Example:

If we apply a **horizontal shear** with $k = 2$ to the point **(3,4)**:

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} (1 \times 3) + (2 \times 4) \\ (0 \times 3) + (1 \times 4) \end{bmatrix}$$

$$= \begin{bmatrix} 3 + 8 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ 4 \end{bmatrix}$$

So, the new point is **(11,4)**.

2. Why Do We Need Transformations?

Vector transformations, including shear, are fundamental in various fields:

A. Computer Graphics

- **Scaling, rotation, and shear** are used to **manipulate images, animations, and 3D models**.
- Example: **Shear transformation is used in text rendering** (italics).

B. Physics & Engineering

- **Shear stress analysis** in materials to understand how forces deform objects.
- Used in **structural engineering** to analyze how buildings react to lateral forces (like wind and earthquakes).

C. Robotics & AI

- Robots use **linear transformations** to move and adjust **arms and joints**.
- Shear transformations are used in **robotic vision systems**.

D. Machine Learning & Data Science

- **Principal Component Analysis (PCA)** applies transformations to reduce data dimensions.

E. Animation & Game Development

- Shear is used to create **realistic movements** and **perspective effects**.

Matrix Multiplication & Composition

1. Matrix Multiplication

Matrix multiplication is a way of combining two matrices to produce a new matrix. It follows a specific set of rules to ensure consistency in transformations, calculations, and operations in various applications like graphics, physics, and engineering.

Rules for Matrix Multiplication

- If matrix **A** has dimensions m×n times and matrix **B** has dimensions n×p, then their product **AB** will be a matrix of size m×p.
- The **number of columns in the first matrix (A)** must match the **number of rows in the second matrix (B)**.

How to Multiply Matrices

Given:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

The product $C = AB$ is calculated as:

$$C = \begin{bmatrix} (a_{11} \cdot b_{11} + a_{12} \cdot b_{21}) & (a_{11} \cdot b_{12} + a_{12} \cdot b_{22}) \\ (a_{21} \cdot b_{11} + a_{22} \cdot b_{21}) & (a_{21} \cdot b_{12} + a_{22} \cdot b_{22}) \end{bmatrix}$$

Example:

Multiply:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$C = AB = \begin{bmatrix} (2 \cdot 1 + 3 \cdot 3) & (2 \cdot 2 + 3 \cdot 4) \\ (1 \cdot 1 + 4 \cdot 3) & (1 \cdot 2 + 4 \cdot 4) \end{bmatrix}$$

$$= \begin{bmatrix} (2 + 9) & (4 + 12) \\ (1 + 12) & (2 + 16) \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 16 \\ 13 & 18 \end{bmatrix}$$

2. Composition of Transformations

What is Composition?

Composition of transformations means **applying multiple transformations in sequence**, where each transformation is represented by a matrix. This is done using **matrix multiplication**.

Why Use Composition?

- Instead of applying multiple transformations **separately**, we can **combine them into a single matrix**.
- This improves **efficiency** in computations (especially in graphics, robotics, and simulations).

Example: Rotation Followed by Scaling

1. **Rotation Matrix by θ:**

Example Calculation

For a **90° rotation** and a **scaling factor of 2**:

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$T = S \cdot R = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (2 \cdot 0 + 0 \cdot 1) & (2 \cdot -1 + 0 \cdot 0) \\ (0 \cdot 0 + 2 \cdot 1) & (0 \cdot -1 + 2 \cdot 0) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$$

Thus, the **combined transformation** rotates and then scales.

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Thus, the **combined transformation** rotates and then scales.

3. Applications for Matrix Multiplication & Composition

- **Computer Graphics:** Combining **rotation, scaling, shear, and translation** for **3D modeling** and animations.
- **Robotics:** Using transformations to move robotic arms **precisely**.
- **Physics & Engineering:** Simulating movements and forces in **mechanical systems**.
- **Data Science:** Feature transformations in **machine learning models**.

Matrix Operations: Addition, Subtraction, Multiplication, Determinant, and Inverse

1. Matrix Addition

Matrix addition is the process of adding two matrices by adding their corresponding elements.

Definition:

If two matrices AAA and BBB have the same dimensions m×n

C=A+B

where each element of C is given by:

Cij=Aij+Bij.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Conditions for Addition:

- Matrices **must have the same dimensions**.

2. Matrix Subtraction

Matrix subtraction is similar to addition but involves subtracting corresponding elements.

Definition:

If A and B are matrices of the same size:

$$C = A - B$$

where each element of C is given by:

$$C_{ij} = A_{ij} - B_{ij}$$

Example:

$$A = \begin{bmatrix} 4 & 6 \\ 8 & 10 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 4-1 & 6-2 \\ 8-3 & 10-4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Conditions for Subtraction:

- Matrices **must have the same dimensions**.

3. Matrix Multiplication

Matrix multiplication is not element-wise but follows specific rules.

Definition:

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, their product $C = AB$ is an $m \times p$ matrix where:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

This means that each element of C is obtained by taking the **dot product of the corresponding row of A with the corresponding column of B**.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} (1 \cdot 5 + 2 \cdot 7) & (1 \cdot 6 + 2 \cdot 8) \\ (3 \cdot 5 + 4 \cdot 7) & (3 \cdot 6 + 4 \cdot 8) \end{bmatrix}$$

$$= \begin{bmatrix} (5 + 14) & (6 + 16) \\ (15 + 28) & (18 + 32) \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Conditions for Multiplication:

- The number of **columns in the first matrix** must equal the **number of rows in the second matrix**.

4. Determinant of a Matrix

The determinant is a scalar value that represents certain properties of a matrix, such as whether it is invertible.

Definition:

For a **2×2 matrix**:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is:

$$\det(A) = ad - bc$$

For a **3×3 matrix**:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Example (2×2 Matrix):

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

$$\det(A) = (3 \cdot 4) - (2 \cdot 1) = 12 - 2 = 10$$

Properties of Determinants:

- If $\det(A) = 0$, the matrix is **singular** and **not invertible**.
- If $\det(A) \neq 0$, the matrix is **invertible**.

5. Inverse of a Matrix

The inverse of a matrix A is denoted as A^{-1} and satisfies:

$$A \cdot A^{-1} = I$$

where I is the identity matrix.

Formula for a 2×2 Matrix:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where $\det(A) \neq 0$.

Example:

For

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

we first find $\det(A)$:

$$\det(A) = (3 \cdot 4) - (2 \cdot 1) = 12 - 2 = 10$$

Now, applying the formula:

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0.4 & -0.2 \\ -0.1 & 0.3 \end{bmatrix}$$

Conditions for Inversion:

- A matrix is **invertible only if** $\det(A) \neq 0$.
- A **singular matrix** (determinant = 0) **has no inverse**.

System of Linear Equations

A **system of linear equations** is a collection of two or more linear equations involving the same set of variables. The goal is to find the values of these variables that satisfy all equations simultaneously.

General Form:

A system of n linear equations with mmm variables (x1, x2,..., xm) can be written as:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = b_n$$

Where:

- aij are coefficients,
- xi are variables,
- bi are constant terms.

Example:

Consider the system:

2x+3y=8

Methods to Solve:

1. **Substitution Method** – Solve one equation for a variable and substitute it into the other equation.
2. **Elimination Method** – Add or subtract equations to eliminate a variable.
3. **Matrix Method** – Represent the system in matrix form and solve using inverse matrices or row operations.

For the given system, solving using elimination:

1. Multiply the second equation by 3:

   12x−3y=6

2. Add to the first equation:

   (2x+3y) +(12x−3y) =8+6

   14x=14

x=1

3. Substitute x=1 into 4x−y=2:

 4(1) −y=2

 4−y=2

 y=2

Solution: x=1, y=2

This means the two equations intersect at the point (1,2).

Types of Liner Equations

Linear equations are classified based on the number of variables they contain. Below are the main types with their definitions and general equations:

1. **Linear Equation in One Variable**
    - An equation that contains only one variable and has a degree of 1.
    - **General Form:** ax+b=0
    - **Example:** 5x−3=7
2. **Linear Equation in Two Variables**
    - An equation that involves two variables and represents a straight line when plotted on a graph.
    - **General Form:** ax+by=c
    - **Example:** 2x+3y=6
3. **Linear Equation in Three Variables**
    - An equation with three variables that represents a plane in three-dimensional space.
    - **General Form:** ax+by+cz=d
    - **Example:** x−2y+3z=5
4. **System of Linear Equations**
    - A set of two or more linear equations involving the same variables.
    - General Form:

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

    - Example:

$$\begin{cases} 2x + y = 5 \\ 3x - 4y = 2 \end{cases}$$

Writing a System of Linear Equations

A system of linear equations can be expressed in different forms, depending on how the information is structured. The three main representations are:

1. **Standard Form**

   - This is the most common form, where each equation is written as a linear combination of variables with their corresponding coefficients.
   - **General Form:**

     a1x+b1y+c1z=d1

     a2x+b2y+c2z=d2

   - Example:

$$2x + 3y = 8$$

$$4x - y = 2$$

2. **Matrix Form**

- The system is written as a matrix equation:
   - AX=B
   - General Form:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

   - Example:

$$\begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

3. **Augmented Matrix Form**

- The system is represented as an augmented matrix, where the constants are included in the last column.

- General Form:

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \end{array}\right]$$

- Example:

$$\left[\begin{array}{cc|c} 2 & 3 & 8 \\ 4 & -1 & 2 \end{array}\right]$$

Graphical Method

The **graphical method** is a way to solve a system of linear equations by plotting them on a coordinate plane. The solution is the point where the lines representing the equations intersect.

Steps to Solve Using the Graphical Method:

1. **Rewrite Equations in Slope-Intercept Form (if necessary)**
   - Convert each equation into the form $y=mx+c$, where m is the slope and c is the y-intercept.
2. **Plot the Lines on a Graph**
   - Identify the y-intercept and plot it.
   - Use the slope to find another point and draw the line.
3. **Finding the Intersection Point**
   - If the lines **intersect at one point**, the system has a unique solution.
   - If the lines are **parallel**, there is **no solution** (inconsistent system).
   - If the lines **coincide**, there are **infinitely many solutions** (dependent system).

Example:

Solve the system using the graphical method:

$$2x+y=8$$
$$x-y=2$$

Step 1: Express in Slope-Intercept Form

1st equation:

$$y=-2x+8$$

2nd equation:

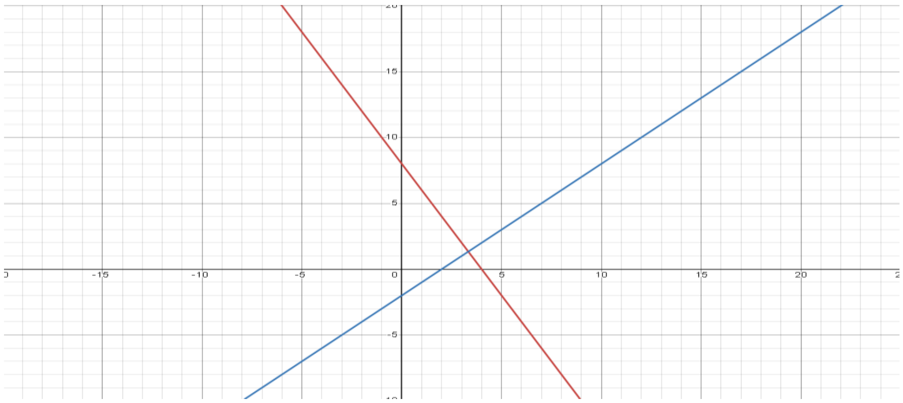
$$y=x-2$$

Step 2: Plot the Lines

- The first equation $y=-2x+8$ has a **y-intercept** at (0,8) and a **slope** of -2.
- The second equation $y=x-2$ has a **y-intercept** at (0, -2) and a **slope** of 1.

Step 3: Find the Intersection

By plotting both lines, they intersect at (2,0) which is the solution to the system.



Substitution and Elimination Method

1. Substitution Method

Definition:

The **substitution method** involves solving one equation for one variable and then substituting that expression into the other equation. This reduces the system to a single equation with one variable, which can be solved easily.

Steps to Solve Using Substitution Method:

1. Solve one equation for one variable in terms of the other.
2. Substitute the expression into the second equation.
3. Solve for the remaining variable.
4. Substitute the found value back into the first equation to get the second variable.

Example:

Solve the system:

$$x + 2y = 10$$

$$3x - y = 5$$

Step 1: Solve for One Variable

Solve the first equation for x:

$$x = 10 - 2y$$

Step 2: Substitute into the Second Equation

$$3(10 - 2y) - y = 5$$

$$30 - 6y - y = 5$$

$$30 - 7y = 5$$

$$-7y = -25$$

$$y = \frac{25}{7}$$

Step 3: Find x

$$x = 10 - 2\left(\frac{25}{7}\right)$$

$$x = 10 - \frac{50}{7}$$

$$x = \frac{70}{7} - \frac{50}{7}$$

$$x = \frac{20}{7}$$

Solution:

$$x = \frac{20}{7}, \quad y = \frac{25}{7}$$

2. Elimination Method

Definition:

The **elimination method** involves adding or subtracting the equations to eliminate one of the variables, reducing the system to a single equation with one variable. This method is also known as the **addition method**.

Steps to Solve Using Elimination Method:

1. Multiply one or both equations (if necessary) to make the coefficients of one variable the same.
2. Add or subtract the equations to eliminate one variable.
3. Solve for the remaining variable.
4. Substitute the found value back into one of the original equations to get the other variable.

Example:

Solve the system:

$$2x + 3y = 12$$

$$4x - y = 5$$

Step 1: Make the Coefficients of One Variable the Same

Multiply the second equation by 3 to match the coefficient of y:

$$2x + 3y = 12$$

$$12x - 3y = 15$$

Step 2: Add the Equations

$$(2x + 3y) + (12x - 3y) = 12 + 15$$

$$14x = 27$$

$$x = \frac{27}{14}$$

Step 3: Substitute x into One of the Original Equations

Using $2x + 3y = 12$:

$$2\left(\frac{27}{14}\right) + 3y = 12$$

$$\frac{54}{14} + 3y = 12$$

$$3y = 12 - \frac{54}{14}$$

$$3y = \frac{168}{14} - \frac{54}{14}$$

$$3y = \frac{114}{14}$$

$$y = \frac{38}{14} = \frac{19}{7}$$

Solution:

$$x = \frac{27}{14}, \quad y = \frac{19}{7}$$

Matrix Inversion Method

Definition:

The **matrix inversion method** is a technique used to solve a system of linear equations using matrices. It involves representing the system as a matrix equation and finding the solution using the inverse of the coefficient matrix.

General Form:

A system of linear equations:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Can be written in **matrix form** as:

$$AX = B$$

Where:

$$A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

The solution is given by:

$$X = A^{-1}B$$

where A^{-1} is the **inverse** of matrix A.

Example:

Solve the system using the matrix inversion method:

$$2x + 3y = 8$$

$$4x - y = 2$$

Step 1: Write the System in Matrix Form

$$A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

$$AX = B$$

Step 2: Find the Inverse of Matrix A

The inverse of a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is given by:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where **det(A)** is the determinant:

$$\det(A) = (2 \times -1) - (3 \times 4) = -2 - 12 = -14$$

Now,

$$A^{-1} = \frac{1}{-14} \begin{bmatrix} -1 & -3 \\ -4 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{14} & \frac{3}{14} \\ \frac{4}{14} & -\frac{2}{14} \end{bmatrix}$$

Perform matrix multiplication:

$$x = \left(\frac{1}{14} \times 8\right) + \left(\frac{3}{14} \times 2\right)$$

$$x = \frac{8}{14} + \frac{6}{14} = \frac{14}{14} = 1$$

$$y = \left(\frac{4}{14} \times 8\right) + \left(-\frac{2}{14} \times 2\right)$$

$$y = \frac{32}{14} - \frac{4}{14} = \frac{28}{14} = 2$$

Solution:

$$x = 1, \quad y = 2$$

Gaussian Elimination Method

Gaussian elimination is a mathematical algorithm used to solve systems of linear equations. It systematically transforms a given system into an equivalent upper triangular or row echelon form using elementary row operations, making it easier to solve for the unknown variables. The method is also used to compute the rank of a matrix and find its inverse.

Steps of Gaussian Elimination

1. **From the Augmented Matrix:** Write the system of linear equations as an augmented matrix.
2. **Forward Elimination:** Convert the matrix into upper triangular form by:
   - Swapping rows if needed.
   - Making leading coefficients (pivot elements) 1.
   - Eliminating coefficients below the pivot using row operations.
3. **Back Substitution:** Solve for the unknown variables by substituting values from the last equation to the first.

Example

Solve the following system of equations using Gaussian elimination:

$$2x + 3y - z = 5$$

$$4x + 7y - 3z = 10$$

$$-2x + 4y + 5z = -2$$

Step 1: Write the Augmented Matrix

$$\begin{bmatrix} 2 & 3 & -1 & |5 \\ 4 & 7 & -3 & |10 \\ -2 & 4 & 5 & |-2 \end{bmatrix}$$

Step 2: Forward Elimination

- Make the first pivot element 1 by dividing the first row by 2:

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 4 & 7 & -3 & |10 \\ -2 & 4 & 5 & |-2 \end{bmatrix}$$

- Eliminate the first column entries below the pivot using row operations:

- $R_2 \to R_2 - 4R_1$
- $R_3 \to R_3 + 2R_1$

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 0 & 1 & -1 & |0 \\ 0 & 7 & 4 & |3 \end{bmatrix}$$

- Make the second pivot 1 (already done), then eliminate below it:
  - $R_3 \to R_3 - 7R_2$

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 0 & 1 & -1 & |0 \\ 0 & 0 & 11 & |3 \end{bmatrix}$$

Step 3: Back Substitution

- From the last row: $11z = 3 \Rightarrow z = \frac{3}{11}$
- From the second row: $y - z = 0 \Rightarrow y = \frac{3}{11}$
- From the first row: $x + 1.5y - 0.5z = 2.5$

$$x + 1.5 \times \frac{3}{11} - 0.5 \times \frac{3}{11} = 2.5$$

$$x + \frac{4.5}{11} = 2.5$$

$$x = 2.5 - \frac{4.5}{11} = \frac{27.5 - 4.5}{11} = \frac{23}{11}$$

Final Solution:

$$x = \frac{23}{11}, \quad y = \frac{3}{11}, \quad z = \frac{3}{11}$$

Gauss-Jordan Elimination Method

The **Gauss-Jordan elimination** method is an extension of **Gaussian elimination** that reduces a given system of linear equations into **reduced row echelon form (RREF)**. Unlike Gaussian elimination, which stops at an upper triangular matrix, Gauss-Jordan elimination continues until the matrix is transformed into the **identity matrix** on the left side.

Steps of Gauss-Jordan Elimination

1. **From the Augmented Matrix** from the system of equations.
2. **Convert to Row Echelon Form** using forward elimination:
    - Use row operations to create leading 1s (pivot elements).
    - Eliminate the values below the pivots.
3. **Convert to Reduced Row Echelon Form** using backward elimination:
    - Eliminate the values above the pivots.
4. **Extract the Solution** from the final matrix.

Example

Solve the system of equations using **Gauss-Jordan Elimination**:

$$2x + 3y - z = 5$$

$$4x + 7y - 3z = 10$$

$$-2x + 4y + 5z = -2$$

Step 1: Write the Augmented Matrix

$$\begin{bmatrix} 2 & 3 & -1 & |5 \\ 4 & 7 & -3 & |10 \\ -2 & 4 & 5 & |-2 \end{bmatrix}$$

Step 2: Convert to Row Echelon Form

- Make the first pivot element **1** by dividing **Row 1** by **2**:

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 4 & 7 & -3 & |10 \\ -2 & 4 & 5 & |-2 \end{bmatrix}$$

- Eliminate the first column below the pivot:
    - $R_2 \to R_2 - 4R_1$
    - $R_3 \to R_3 + 2R_1$

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 0 & 1 & -1 & |0 \\ 0 & 7 & 4 & |3 \end{bmatrix}$$

- Make the second pivot **1** (already done), then eliminate below and above it:
  - $R_3 \rightarrow R_3 - 7R_2$

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 0 & 1 & -1 & |0 \\ 0 & 0 & 11 & |3 \end{bmatrix}$$

Step 3: Convert to Reduced Row Echelon Form

- Make the third pivot **1** by dividing **Row 3** by **11**:

$$\begin{bmatrix} 1 & 1.5 & -0.5 & |2.5 \\ 0 & 1 & -1 & |0 \\ 0 & 0 & 1 & |\frac{3}{11} \end{bmatrix}$$

- Eliminate the third column entries above using:
  - $R_2 \rightarrow R_2 + R_3$
  - $R_1 \rightarrow R_1 + 0.5R_3$

$$\begin{bmatrix} 1 & 1.5 & 0 & |2.5 + 0.5 \times \frac{3}{11} \\ 0 & 1 & 0 & |\frac{3}{11} \\ 0 & 0 & 1 & |\frac{3}{11} \end{bmatrix}$$

- Make the second pivot 1 and adjust:

$$\begin{bmatrix} 1 & 0 & 0 & |\frac{23}{11} \\ 0 & 1 & 0 & |\frac{3}{11} \\ 0 & 0 & 1 & |\frac{3}{11} \end{bmatrix}$$

Step 4: Extract the Solution

$$x = \frac{23}{11}, \quad y = \frac{3}{11}, \quad z = \frac{3}{11}$$

LU Decomposition Method

LU Decomposition (also called **LU Factorization**) is a method used to solve systems of linear equations by breaking a matrix into two simpler matrices:
A=LUA = LUA=LU
Where:

- LLL (Lower triangular matrix) has 1s on the diagonal and values below the diagonal.
- UUU (Upper triangular matrix) has values above the diagonal.

Why Use LU Decomposition?

- It simplifies solving multiple systems of equations with the same coefficient matrix.
- It's computationally efficient for large matrices.
- It is used in numerical methods like **determinant calculation and inverse matrix finding**.

Steps for LU Decomposition

1. **Write the system as a matrix equation:** Ax=b.
2. **Factorize** AAA into the product of a lower triangular matrix L and an upper triangular matrix U, i.e., A=LU.
3. **Solve Ly=b** (Forward substitution).
4. **Solve Ux=y** (Backward substitution) to get the final solution.

Example

Solve the system using **LU Decomposition**:

$$2x + 3y + z = 5$$

$$4x + 7y + 3z = 10$$

$$-2x + 4y + 5z = -2$$

Step 1: Write the Augmented Matrix

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 3 \\ -2 & 4 & 5 \end{bmatrix}$$

Step 2: Perform LU Decomposition

We want to find **L** and **U**, where:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

1. The first row of U is the same as the first row of A:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

2. Find l_{21} and l_{31}:

$$l_{21} = \frac{4}{2} = 2, \quad l_{31} = \frac{-2}{2} = -1$$

   So,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & l_{32} & 1 \end{bmatrix}$$

3. Modify U using row operations:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 + R_1$$

   Gives:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 7 & 6 \end{bmatrix}$$

4. Find l_{32}:

$$l_{32} = \frac{7}{1} = 7$$

   So,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 7 & 1 \end{bmatrix}$$

5. Modify U further:

$$R_3 \to R_3 - 7R_2$$

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

Thus, the LU decomposition of A is:

$$A = LU$$

where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 7 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

Step 3: Solve $Ly = b$ (Forward Substitution)

We solve:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 7 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ -2 \end{bmatrix}$$

Solving step by step:

$$y_1 = 5$$

$$2y_1 + y_2 = 10 \Rightarrow 2(5) + y_2 = 10 \Rightarrow y_2 = 0$$

$$-1y_1 + 7y_2 + y_3 = -2 \Rightarrow -5 + 7(0) + y_3 = -2 \Rightarrow y_3 = 3$$

So,

$$y = \begin{bmatrix} 5 \\ 0 \\ 3 \end{bmatrix}$$

Step 4: Solve $Ux = y$ (Backward Substitution)

We solve:

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 3 \end{bmatrix}$$

Solving step by step:

$$-1x_3 = 3 \Rightarrow x_3 = -3$$

$$x_2 + x_3 = 0 \Rightarrow x_2 - 3 = 0 \Rightarrow x_2 = 3$$

$$2x_1 + 3x_2 + x_3 = 5$$

$$2x_1 + 3(3) + (-3) = 5$$

$$2x_1 + 9 - 3 = 5$$

$$2x_1 = -1 \Rightarrow x_1 = -\frac{1}{2}$$

Final Solution:

$$x_1 = -\frac{1}{2}, \quad x_2 = 3, \quad x_3 = -3$$

Singular Value Decomposition (SVD) Method

Singular Value Decomposition (SVD) is a **factorization technique** used in linear algebra for decomposing a matrix into three simpler matrices. It is widely used in **data compression, image processing, machine learning, and numerical computing**.

Definition

For a given **m × n** matrix A, the **SVD** is given by:

$A=U\Sigma V^T$

Where:

- U is an **m × m orthogonal matrix** (left singular vectors).
- Σ is an **m × n diagonal matrix** with non-negative values (**singular values**).
- V^T is an **n × n orthogonal matrix** (right singular vectors).

The **singular values** in Σ represent the magnitude of variation in the data along different dimensions.

Example of SVD

Consider a **2 × 2** matrix:

$$A = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix}$$

Step 1: Compute A^TA and AA^T

We first compute:

$$A^TA = \begin{bmatrix} 4 & 3 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 16+9 & 0+(-15) \\ 0+(-15) & 0+25 \end{bmatrix} = \begin{bmatrix} 25 & -15 \\ -15 & 25 \end{bmatrix}$$

And,

$$AA^T = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} 16 & 12 \\ 12 & 34 \end{bmatrix}$$

Step 2: Find Eigenvalues of A^TA

To find the singular values, we solve:

$$\det(A^TA - \lambda I) = 0$$

$$\begin{vmatrix} 25-\lambda & -15 \\ -15 & 25-\lambda \end{vmatrix} = 0$$

Expanding,

$$(25 - \lambda)(25 - \lambda) - (-15)(-15) = 0$$

$$(25 - \lambda)^2 - 225 = 0$$

$$(25 - \lambda) = \pm 15$$

Solving for λ,

$$\lambda = 40, 10$$

Since **singular values** are the **square roots of eigenvalues**, we get:

$$\sigma_1 = \sqrt{40}, \quad \sigma_2 = \sqrt{10}$$

Step 3: Compute U, Σ, and V

Using eigenvectors, we compute:

$$U = [\text{Left singular vectors}]$$

$$V = [\text{Right singular vectors}]$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

Final Decomposition:

$$A = U \Sigma V^T$$

Step 1: Compute the Right Singular Vectors (Columns of V)

The **right singular vectors** V are the **eigenvectors** of $A^T A$.

We already computed:

$$A^T A = \begin{bmatrix} 25 & -15 \\ -15 & 25 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = 40$ and $\lambda_2 = 10$, corresponding to **singular values** $\sigma_1 = \sqrt{40}$ and $\sigma_2 = \sqrt{10}$.

Find Eigenvectors for $\lambda_1 = 40$

Solve:

$$(A^T A - 40I)v = 0$$

$$\begin{bmatrix} 25 - 40 & -15 \\ -15 & 25 - 40 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -15 & -15 \\ -15 & -15 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

Solving $-15v_1 - 15v_2 = 0$, we get $v_1 = -v_2$.
Choosing $v_1 = \frac{1}{\sqrt{2}}$, we get:

$$v_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Find Eigenvectors for $\lambda_2 = 10$

Solve:

$$(A^T A - 10I)v = 0$$

$$\begin{bmatrix} 25-10 & -15 \\ -15 & 25-10 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 15 & -15 \\ -15 & 15 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

Solving $15v_1 - 15v_2 = 0$, we get $v_1 = v_2$.
Choosing $v_1 = \frac{1}{\sqrt{2}}$, we get:

$$v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Thus, the matrix V is:

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 2: Compute the Left Singular Vectors (Columns of U)

The **left singular vectors** are the **eigenvectors** of AA^T.

We computed:

$$AA^T = \begin{bmatrix} 16 & 12 \\ 12 & 34 \end{bmatrix}$$

Since AA^T has the same eigenvalues as $A^T A$, we use $\lambda_1 = 40$, $\lambda_2 = 10$ to find eigenvectors.

Find Eigenvectors for $\lambda_1 = 40$

Solve:

$$(AA^T - 40I)u = 0$$

$$\begin{bmatrix} 16-40 & 12 \\ 12 & 34-40 \end{bmatrix} = \begin{bmatrix} -24 & 12 \\ 12 & -6 \end{bmatrix}$$

Solving $-24u_1 + 12u_2 = 0 \Rightarrow u_1 = \frac{1}{\sqrt{5}}, u_2 = \frac{2}{\sqrt{5}}$, we get:

$$u_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

Find Eigenvectors for $\lambda_2 = 10$

Solve:

$$(AA^T - 10I)u = 0$$

$$\begin{bmatrix} 16-10 & 12 \\ 12 & 34-10 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 12 & 24 \end{bmatrix}$$

Solving $6u_1 + 12u_2 = 0 \Rightarrow u_1 = -2u_2$, we get:

$$u_2 = \begin{bmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

Thus, the matrix U is:

$$U = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

Step 3: Construct Σ

$$\Sigma = \begin{bmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2\sqrt{10} & 0 \\ 0 & \sqrt{10} \end{bmatrix}$$

Final SVD Decomposition

$$A = U\Sigma V^T$$

Where:

$$U = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2\sqrt{10} & 0 \\ 0 & \sqrt{10} \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are fundamental concepts in linear algebra that help in understanding the transformation properties of matrices. They are widely used in **machine**

learning, quantum mechanics, vibration analysis, image processing, and principal component analysis (PCA).

1. Definition

For a given **square matrix** AAA, an **eigenvector** v and an **eigenvalue** λ satisfy the equation:

Av=λv

Where:

- A is an n×n matrix.
- v is a **nonzero vector** (eigenvector).
- λ is a **scalar** (eigenvalue).

The eigenvector remains in the same direction after transformation by AAA, only scaled by λ.

2. How to Compute Eigenvalues and Eigenvectors

Step 1: Find the Eigenvalues

Eigenvalues are found by solving the **characteristic equation**:

det(A−λI) =0

Where III is the identity matrix.

Step 2: Find the Eigenvectors

For each eigenvalue λ, solve:

(A−λI) v=0

to find the corresponding eigenvector v.

3. Example Calculation

Let's take a simple **2×2 matrix**:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

Step 1: Find the Eigenvalues

Solve $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

Expanding the determinant:

$$(4-\lambda)(3-\lambda) - (2 \times 1) = 0$$

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

Factorizing:

$$(\lambda - 5)(\lambda - 2) = 0$$

So, the **eigenvalues** are:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

Step 2: Find the Eigenvectors

For each eigenvalue, solve $(A - \lambda I)v = 0$.

For $\lambda_1 = 5$:

$$(A - 5I) = \begin{bmatrix} 4-5 & 2 \\ 1 & 3-5 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$$

Solving:

$$\begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $-x + 2y = 0 \Rightarrow x = 2y$.
Choosing $y = 1$, we get $x = 2$.

So, the eigenvector for $\lambda_1 = 5$ is:

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 2$:

$$(A - 2I) = \begin{bmatrix} 4-2 & 2 \\ 1 & 3-2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$$

Solving:

$$\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $2x + 2y = 0 \Rightarrow x = -y$.
Choosing $y = 1$, we get $x = -1$.

So, the eigenvector for $\lambda_2 = 2$ is:

$$v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

4. Final Answer

The **eigenvalues** and **eigenvectors** of A are:

Eigenvalues:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

Eigenvectors:

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$
